

Project Design Phase
Proposed Solution Template

Date	8 July 2026
Team ID	SWTID-2026-5356
Project Name	India's Agricultural Crop Production Analysis
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No	Parameter	Description
1.	Problem Statement (Problem to be solved)	Agricultural crop production varies significantly across different states, districts, seasons, and years. Farmers, researchers, and government organizations often lack a simple way to analyze production trends, cultivated area, and crop performance, making it difficult to plan resources and improve agricultural productivity.
2.	Idea / Solution description	The proposed solution uses the India Agricultural Crop Production (1997-2021) dataset containing crop production records from different states and districts. An interactive Tableau Dashboard and Tableau Story are developed to visualize crop-wise production, seasonal trends, state-wise and district-wise analysis, cultivated area, and yearly production changes. These dashboards are integrated into a Flask-based web application, allowing users to explore agricultural insights using interactive filters and charts through a single user-friendly interface.
3.	Novelty / Uniqueness	Instead of using static reports, the proposed solution provides an interactive analytics platform where users can filter data by state, district, crop, season, and year to generate real-time insights. Integrating Tableau dashboards with a Flask web application makes agricultural data analysis simple, accessible, and useful for decision-making by farmers, researchers, and policymakers.
4.	Social Impact / Customer Satisfaction	The solution helps farmers, agricultural officers, researchers, and government agencies understand crop production patterns and identify high-performing crops and regions. Better insights support improved crop planning, efficient resource allocation, and informed agricultural policies, ultimately increasing productivity and farmer satisfaction.

5.	Business Model (Revenue Model)	The solution can be offered as a subscription-based agricultural analytics platform for government departments, agricultural organizations, research institutes, and agribusiness companies. Additional revenue can be generated through premium dashboards, customized reports, data analytics consulting, and decision-support services for agricultural planning.
6.	Scalability of the Solution	The solution can be expanded by incorporating real-time agricultural data such as weather conditions, rainfall, soil quality, fertilizer usage, and market prices. It can also support predictive analytics using machine learning to forecast crop production and can be extended to cover additional regions, crops, and future datasets without major changes to the existing system.